

04 Determinants



4.3 Common Properties of Determinants

Six common properties of determinants, understood through geometric intuition

Linear Algebra Made Easy and Visual | Open-source code & drafts: <https://github.com/Visualize-ML>

Core Skills for This Section

1

When a square matrix's column vectors are collinear or coplanar, the area/volume is 0, so the matrix is not invertible

3

Swapping any two rows or two columns negates the determinant

5

A permutation matrix, used to swap rows or columns, is orthogonal with determinant ± 1

7

Behind transpose-invariance lies the singular value decomposition (SVD): $\mathbf{A} = \mathbf{U}\mathbf{S}\mathbf{V}^T$

2

The determinant of a matrix product equals the product of the determinants: $\det(\mathbf{AB}) = \det(\mathbf{A}) \cdot \det(\mathbf{B})$

4

Scaling every row or column of a matrix by a scalar k scales the determinant by k^n

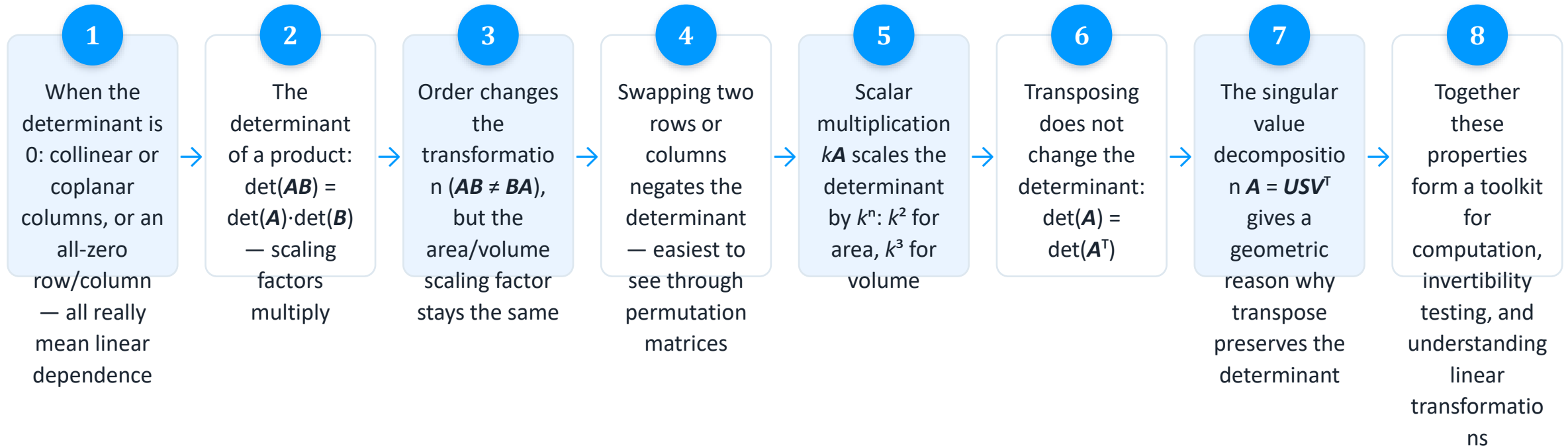
6

Transposing a square matrix does not change its determinant: $\det(\mathbf{A}) = \det(\mathbf{A}^T)$

8

Together these properties form a toolkit for testing invertibility and understanding linear transformations

Roadmap for This Section



Case One: Collinear Column Vectors

$$A = \begin{bmatrix} 1 & 3 \\ 1 & 3 \end{bmatrix} \Rightarrow \det(A) = 0$$

- A square matrix's determinant can be 0 in several ways — the following examples walk through each one
- When $\mathbf{a}_1$ and $\mathbf{a}_2$ lie on the same line, the parallelogram they span has area 0
- The 3×3 matrix $\begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 1 & 2 & 3 \end{bmatrix}$ has three collinear columns too — they can't span a parallelepiped, so the volume is 0

Figure 1 · Collinear Column Vectors

- (a) In 2D: $\mathbf{a}_1$ and $\mathbf{a}_2$ are collinear, so the parallelogram they'd span collapses into a line segment — area 0
- (b) In 3D: three collinear column vectors can't span a parallelepiped — volume 0

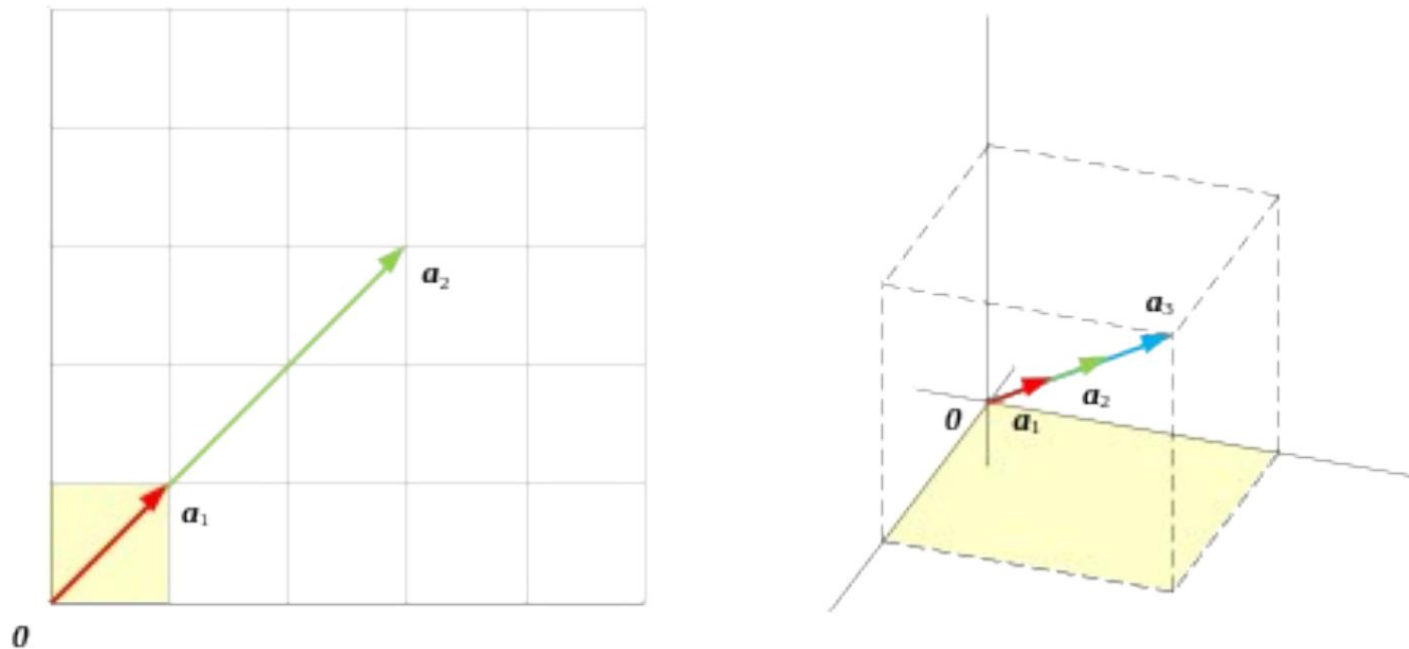


图 1. 矩阵的列向量共线

Case Two: Coplanar Column Vectors

$$A = [3 \ 0 \ 3; 0 \ 3 \ 3; 0 \ 0 \ 0] \Rightarrow \det(A) = 0$$

- This 3×3 matrix's three columns aren't collinear, but they all lie in the same plane — again unable to span a parallelepiped
- Whenever the column vectors have a linear-combination relationship (one can be written as a combination of the others), the determinant is 0

Figure 2 · Coplanar Columns of a 3×3 Matrix

- The red $\mathbf{a}_1$, green $\mathbf{a}_2$, and blue $\mathbf{a}_3$ all lie on the same yellow plane — no solid parallelepiped can form
- The volume (determinant) is 0; this kind of matrix is singular and not invertible

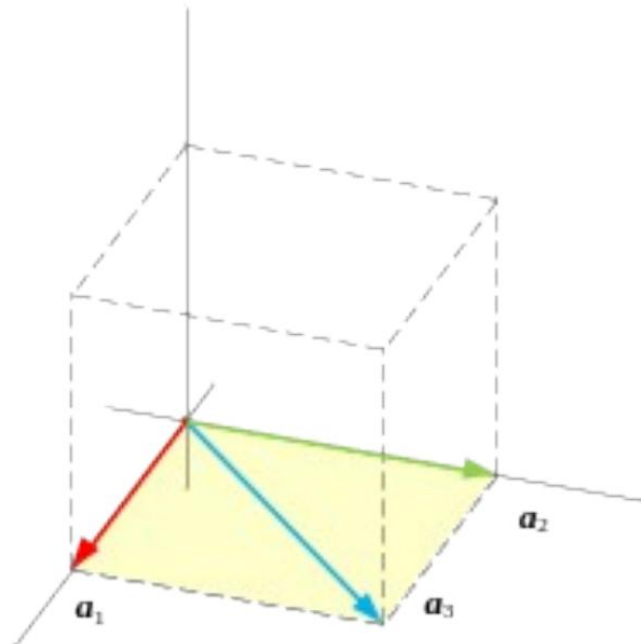


图 2. 3×3 矩阵的列向量共面

Case Three: An All-Zero Row or Column

$$\det([0 \ 2; 0 \ 2]) = 0, \quad \det([0 \ 0; 2 \ 2]) = 0$$

- If a matrix has an entire row or column of zeros, its determinant must be 0 — an extreme case of collinearity/coplanarity
- In summary: whenever a square matrix $\mathbf{A}$'s determinant is 0, its column vectors (and row vectors) are linearly dependent, and $\mathbf{A}$ is a singular, non-invertible matrix

$$\det(\mathbf{AB}) = \det(\mathbf{A}) \cdot \det(\mathbf{B})$$

$$\det(\mathbf{AB}) = \det(\mathbf{A}) \cdot \det(\mathbf{B})$$

- The determinant can be understood as an area (or volume) scaling factor: $\mathbf{A}$ scales area by $\det(\mathbf{A})$, $\mathbf{B}$ scales it by $\det(\mathbf{B})$
- Applying two transformations in sequence multiplies their scaling factors together — that's exactly what $\det(\mathbf{AB})$ means geometrically
- This property holds for square matrices of any dimension, and is one of the core multiplicative properties of the determinant

$\mathbf{AB} \neq \mathbf{BA}$, but $\det(\mathbf{AB}) = \det(\mathbf{BA})$

$$\mathbf{A} = \begin{bmatrix} 1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & 1/2 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} -2 & 0 \\ 0 & -1/2 \end{bmatrix} \Rightarrow \det(\mathbf{A}) = \det(\mathbf{B}) = 1$$

- $\det(\mathbf{AB}) = \det(\mathbf{BA}) = \det(\mathbf{A}) \cdot \det(\mathbf{B})$ — the order of multiplication doesn't affect the total area-scaling factor
- But $\mathbf{AB} \neq \mathbf{BA}$: the resulting shape and orientation differ depending on the order, even though the area comes out the same
- $\mathbf{A}$ rotates the plane about the origin, $\mathbf{B}$ “scales + mirrors” it; the next two slides show the geometric transformations of $\mathbf{ABx}$ and $\mathbf{BAx}$

Figure 3 · The Geometric Transformation of $ABx = y$

- The unit square from column vector x has area 1; after B (scale + mirror) then A (rotate), the area is still 1
- B squashes and mirrors the square into a thin rectangle, and A then rotates it into its final orientation

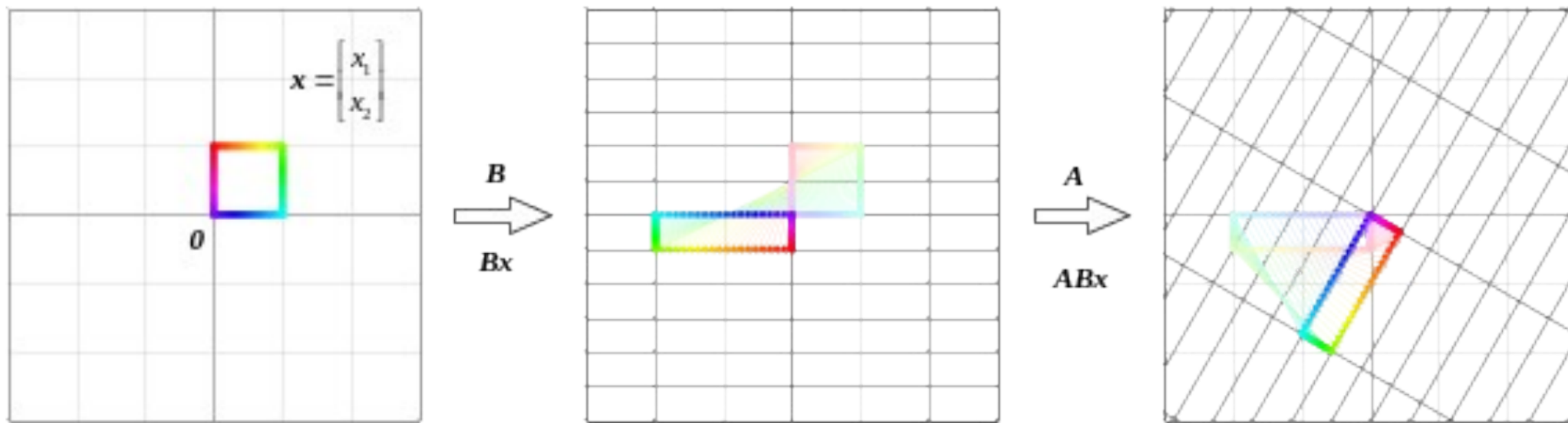


图 3. $ABx = y$ 对应的几何变换

Figure 4 · The Geometric Transformation of $B\mathbf{A}\mathbf{x} = \mathbf{y}$

- First $\mathbf{A}$ (rotate), then $\mathbf{B}$ (scale + mirror): the final shape differs from Figure 3, but its area is still 1
- This visually demonstrates that “ $\mathbf{AB} \neq \mathbf{BA}$, yet $\det(\mathbf{AB}) = \det(\mathbf{BA})$ ”

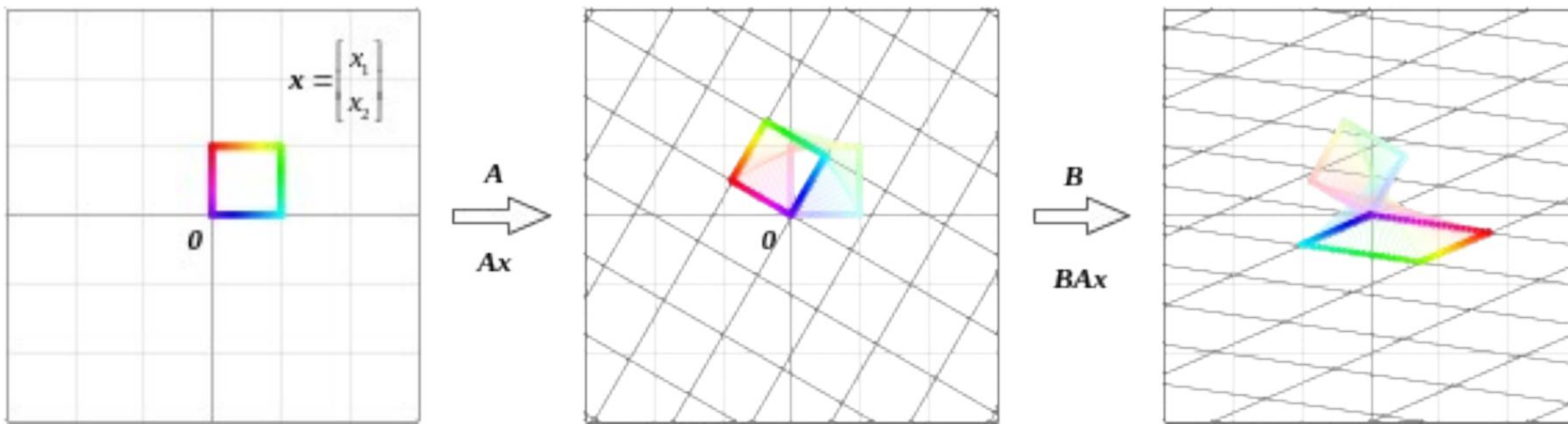


图 4. $B\mathbf{A}\mathbf{x} = \mathbf{y}$ 对应的几何变换

Swapping Any Two Rows or Columns Negates the Determinant

A permutation matrix: a special square matrix obtained by permuting the rows or columns of the identity matrix

- If a matrix's determinant is nonzero, swapping any two of its columns negates the determinant; swapping any two rows does the same
- A permutation matrix's entries are only 0s and 1s, with exactly one 1 in every row and column; a permutation matrix is orthogonal
- Take the 2×2 permutation matrix $\mathbf{P} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ as an example: $\det(\mathbf{P}) = -1$

AP Swaps Columns, PA Swaps Rows

$A = [a \ b; c \ d], P = [0 \ 1; 1 \ 0] \Rightarrow AP$ swaps A 's two columns; PA swaps A 's two rows

- In more elaborate examples (say, a 4×4 matrix), a permutation matrix can swap just two specific columns or rows while leaving the rest unchanged
- Each column (or row) swap flips the determinant's sign once; an even number of swaps leaves the sign unchanged, an odd number flips it

SCALAR MULTIPLICATION

$$\det(k\mathbf{A}) = k^n \cdot \det(\mathbf{A})$$

For an $n \times n$ matrix $\mathbf{A}$, scalar multiplication $k\mathbf{A}$ has determinant $\det(k\mathbf{A}) = k^n \cdot \det(\mathbf{A})$

- $k\mathbf{A}$ multiplies every row (or column) of $\mathbf{A}$ by k — geometrically, every coordinate axis is scaled by k
- A 2×2 matrix corresponds to area, so scaling by k scales both length and width, giving k^2 total; a 3×3 matrix corresponds to volume, scaling by k^3 , and so on

Figure 5 · Scalar 2 in $2A$ for a 2×2 Matrix A

- The unit square from $\mathbf{x}$ stays a unit square (area 1) after the transformation A
- Scalar 2 doubles every edge length of the plane figure, scaling area by $2^2 = 4$

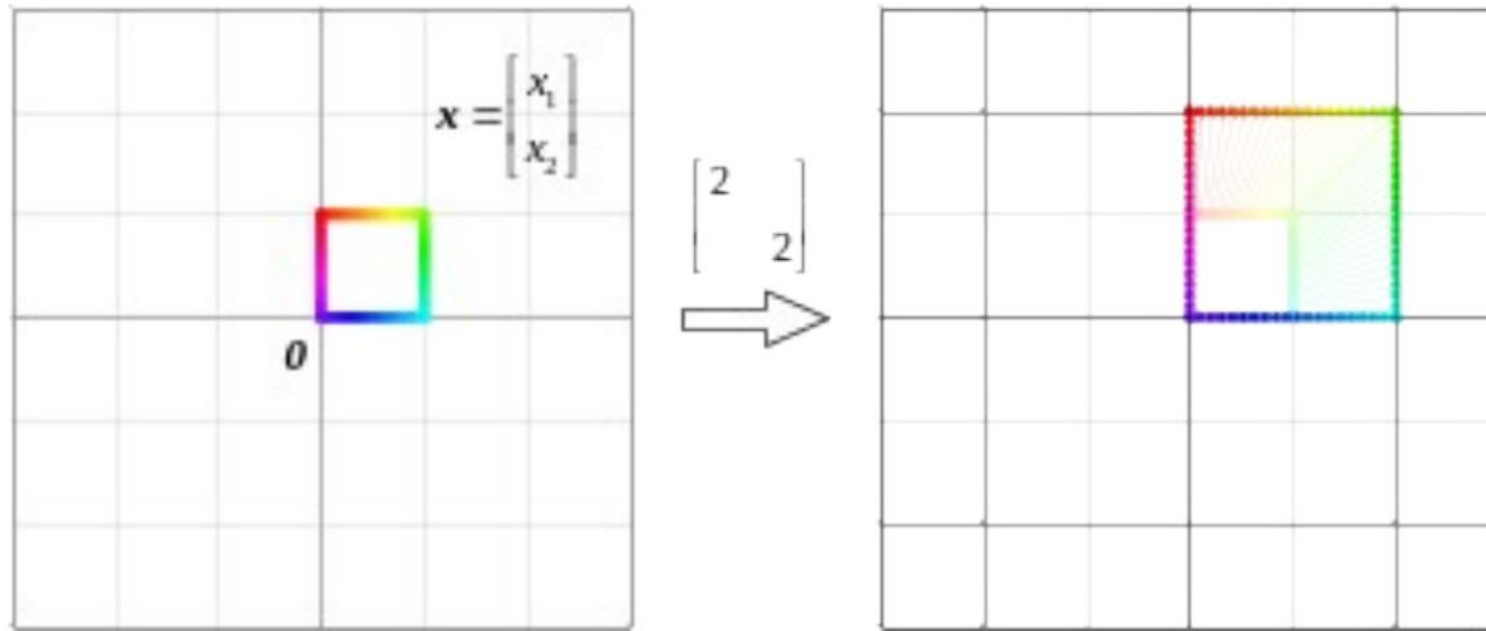


图 5. 给定 2×2 矩阵 A , $2A$ 中标量 2 对应几何变换

Figure 6 · Scalar 2 in $2A$ for a 3×3 Matrix A

- In 3D, scalar 2 doubles the length of every edge
- The volume (determinant) scales by $2^3 = 8$, matching $\det(2A) = 2^3 \cdot \det(A)$

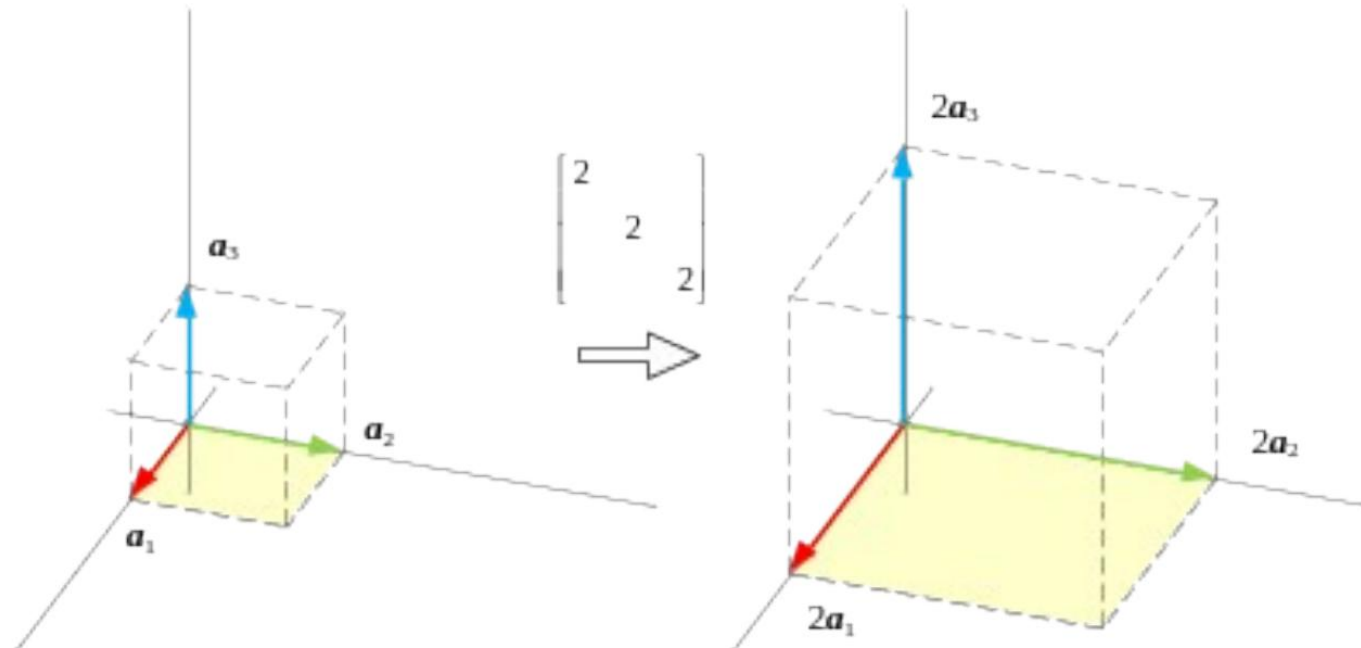


图 6. 给定 3×3 矩阵 A , $2A$ 中标量 2 对应几何变换

A Matrix and Its Transpose Share the Same Determinant

$$\det(\mathbf{A}) = \det(\mathbf{A}^T)$$

- A square matrix $\mathbf{A}$'s determinant equals the determinant of its transpose $\mathbf{A}^T$
- Explaining this geometrically requires the singular value decomposition (SVD)

Singular Value Decomposition: Rotate · Scale · Rotate

$$A = USV^T$$

- U is an orthogonal matrix representing a rotation, with $|\det(U)| = 1$
- S is a diagonal matrix representing scaling; clearly $S = S^T$
- V is orthogonal, representing a rotation; V^T is orthogonal too, with $|\det(V)| = 1$

Given Matrix $\mathbf{A}$, $\det(\mathbf{A}) = 6$

$$\mathbf{A} = \begin{bmatrix} -1.2 & -2.4 \\ 2.4 & -0.2 \end{bmatrix} \Rightarrow \det(\mathbf{A}) = 6$$

- Figure 7 shows the “all-at-once” geometric transformation matrix $\mathbf{A}$ applies to column vector $\mathbf{x}$
- Replacing $\mathbf{A}$ with $\mathbf{USV}^T\mathbf{x}$ gives an equivalent step-by-step transformation, shown in Figure 8

Figure 7 · The All-at-Once Transformation $Ax = y$

- The unit square, transformed by A , lands directly on the final parallelogram
- This is A 's overall effect on x — it doesn't reveal how much rotation versus scaling contributed

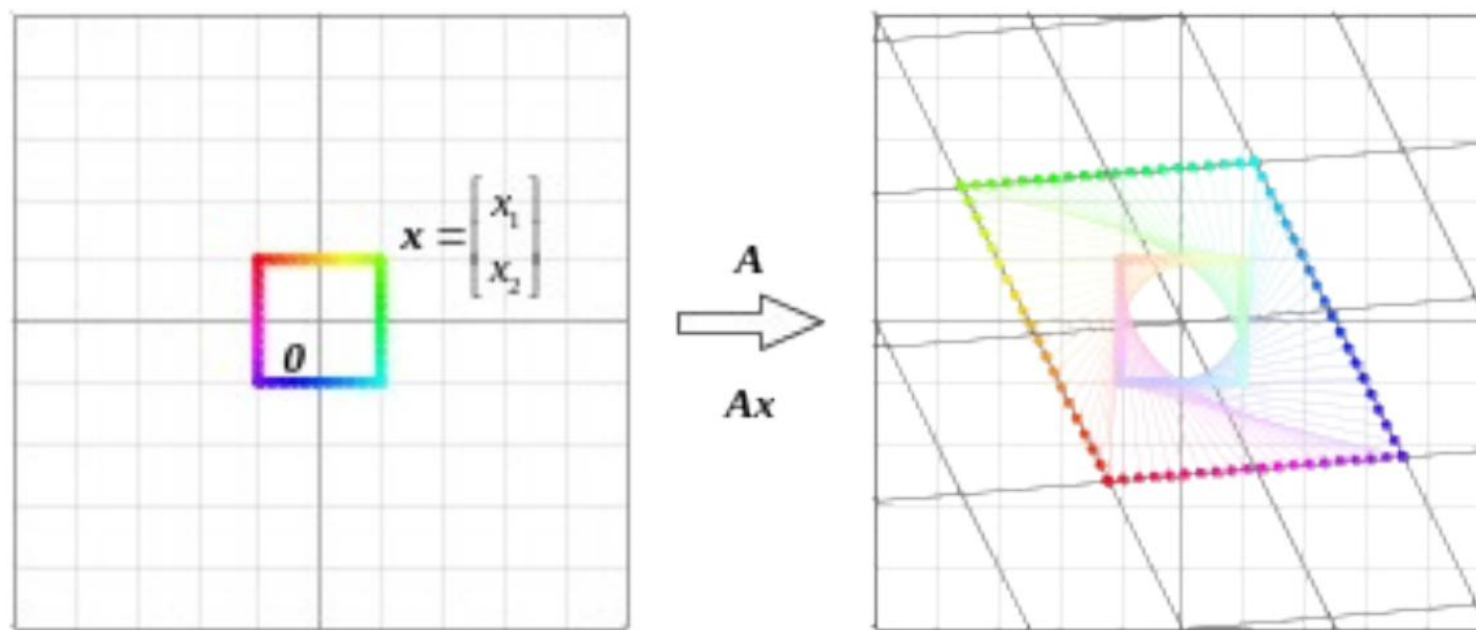
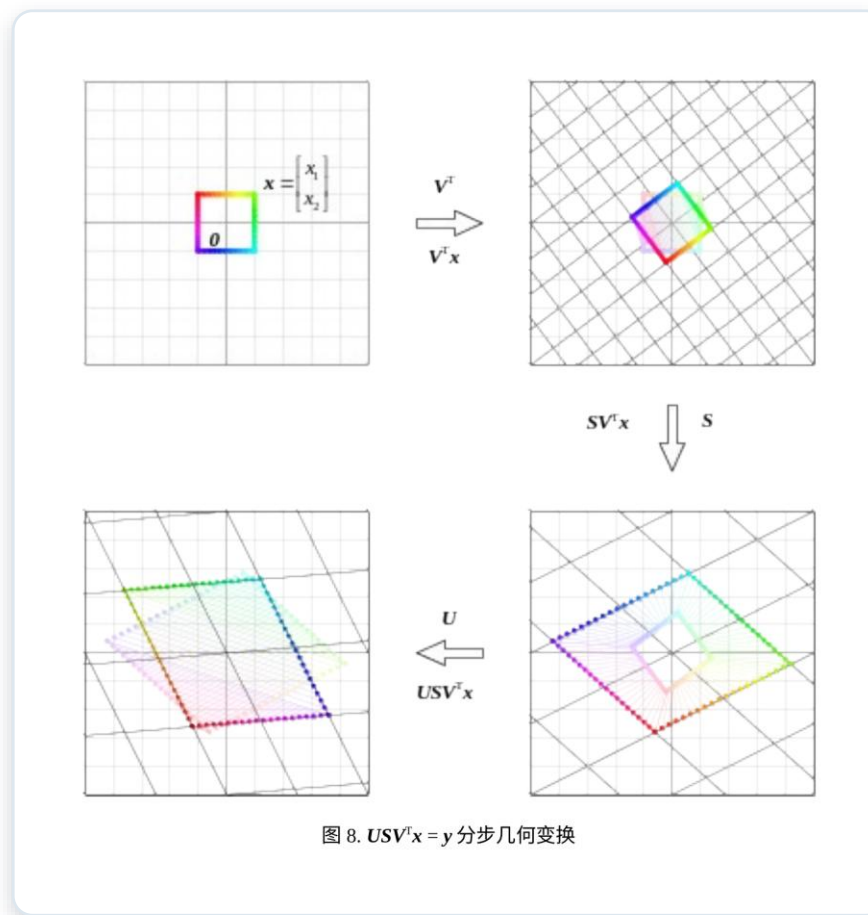


图 7. $Ax = y$ 一步到位的几何变换

Figure 8 · Step-by-Step: Rotate, Scale, Rotate

- First rotate by V^T , then scale along the axes by S , then rotate again by U — the combined effect exactly matches Figure 7
- Only the S step actually changes area; both rotation steps preserve it ($|\det| = 1$)



Decomposing the Transpose of $\mathbf{A}$

$$(\mathbf{A})^T = (\mathbf{USV}^T)^T = \mathbf{VS}^T\mathbf{U}^T = \mathbf{VSU}^T$$

- Geometrically, the factor that truly governs a figure's area or volume in $\mathbf{A}$ is the diagonal matrix $\mathbf{S}$
- Both $\mathbf{A}$ and $\mathbf{A}^T$ share that same $\mathbf{S}$ in their decomposition — which tells us $\mathbf{Ax}$ and $\mathbf{A}^T\mathbf{x}$ produce exactly the same area/volume change
- This is the geometric reason $\det(\mathbf{A}) = \det(\mathbf{A}^T)$ holds

Figure 9 · The All-at-Once Transformation $A^T x = y$

- The overall geometric effect of A^T on x looks different in shape from Ax in Figure 7, but the area is again 6
- Both share the same det value, confirming $\det(A) = \det(A^T)$

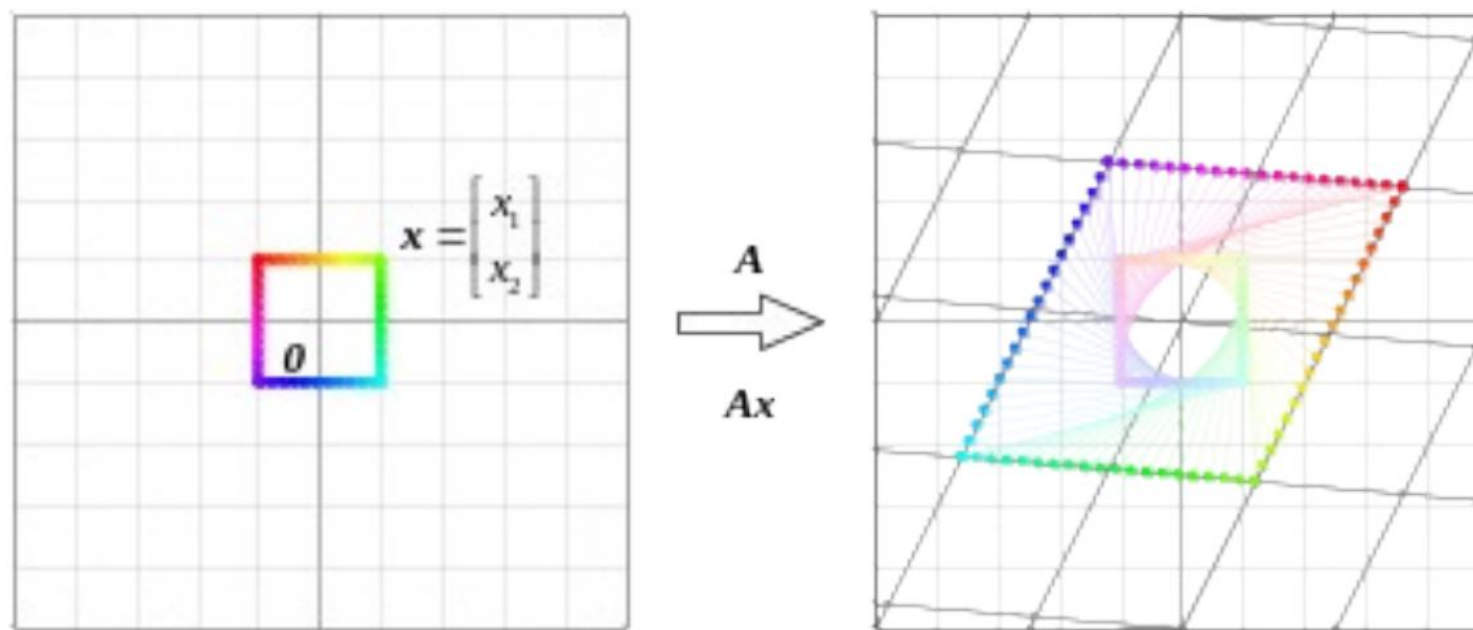
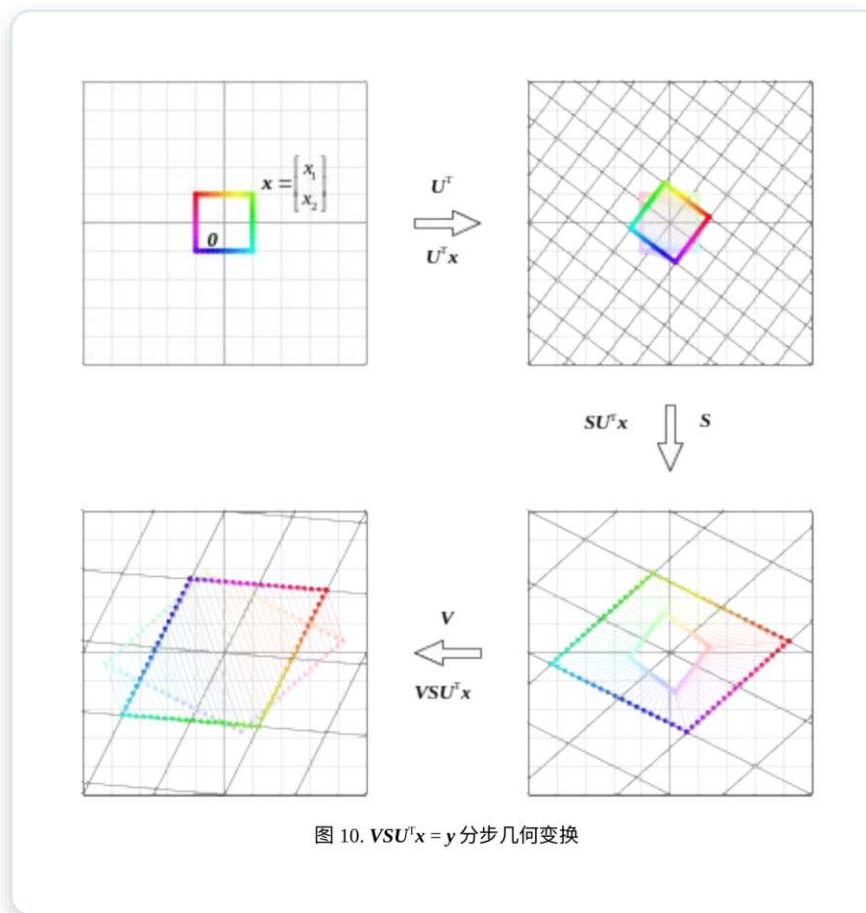


图 9. $A^T x = y$ 一步到位的几何变换

Figure 10 · Step-by-Step, Applied to the Transpose

- The step-by-step path for $A^T = VSU^T$: first rotate by U^T , then scale by S , then rotate by V
- Comparing Figure 8 and Figure 10: both paths use the very same S , only the two rotation matrices swap places — that's the geometric essence of transposing



Six Common Properties, One Geometric Intuition

*collinear/coplanar $\Rightarrow 0$; $\det(AB)=\det(A)\det(B)$; swap rows/cols flips sign;
 $\det(kA)=k^n\det(A)$; $\det(A)=\det(A^T)$*

- $\det = 0 \Leftrightarrow$ linearly dependent columns (collinear, coplanar, or an all-zero row/column) $\Leftrightarrow$ the matrix is not invertible
- The determinant is multiplicative: $\det(\mathbf{AB}) = \det(\mathbf{A}) \cdot \det(\mathbf{B})$; swapping two rows or columns flips the sign; scalar multiplication scales by k^n
- Transposing preserves the determinant, and SVD gives this a clean geometric explanation: $\mathbf{A}$ and $\mathbf{A}^T$ share the very same scaling matrix $\mathbf{S}$

Key Terms in This Section

Linear Dependence *columns collinear, coplanar, or with an all-zero row/column — the determinant is 0*

$\det(AB)=\det(A)\det(B)$ *the determinant of a product equals the product of the determinants*

Row/Column Swap Flips Sign *swapping any two rows or columns of a square matrix negates its determinant*

Transpose Invariance *$\det(A) = \det(A^T)$ — a square matrix and its transpose share the same determinant*

Orthogonal Matrix *a matrix satisfying $QQ^T=I$; represents a rotation or reflection, with $|\det| = 1$*

Singular Matrix *a square matrix with determinant 0 — not invertible*

Permutation Matrix *obtained by permuting the identity matrix's rows/columns; orthogonal, with determinant ± 1*

$\det(kA)=k^n\det(A)$ *scalar multiplication kA scales the determinant by k^n , where n is the matrix's dimension*

SVD *Singular Value Decomposition — $A = USV^T$, with U, V orthogonal and S diagonal*

The Diagonal Matrix S *the scaling matrix in SVD, satisfying $S = S^T$ — the only part that changes area/volume*

GO FURTHER

Companion Code and Further Reading

Further Reading · `numpy.linalg.svd`

<https://numpy.org/doc/2.2/reference/generated/numpy.linalg.svd.html> — the NumPy function for singular value decomposition

Further Reading · Singular Value Decomposition (SVD)

A later chapter of this book covers SVD in full — for now, just get comfortable with $A = USV^T$

Coming Up · Next Section

4.4 Computing Determinants by Hand: a systematic walkthrough of manual techniques for 2×2 , 3×3 , and higher-order determinants

Open-Source Repository

Code and figures for the whole book: <https://github.com/Visualize-ML>

Questions Q1 – Q7

Q1 Determine whether each determinant below is 0: ① $[1, -2; 2, -4]$; ② $[1, -2, 0; 2, 1, 1; 0, 0, 0]$; ③ $[0, -1, 1; 1, -1, 0; 0, 0, 0]$.

Q3 Given any 3×3 matrix A and a permutation matrix P , explain the geometric transformation AP and PA each perform: ① $P = [0, 0, 1; 0, 1, 0; 1, 0, 0]$; ② $P = [1, 0, 0; 0, 0, 1; 0, 1, 0]$; ③ $P = [0, 0, 1; 1, 0, 0; 0, 1, 0]$.

Q5 Learn to use `numpy.linalg.svd` to decompose the matrix $A = [-1.2, -2.4; 2.4, -0.2]$ from this section, and verify the determinant of each of U , S , and V .

Q7 Construct 5 sets of 2×2 matrices with collinear columns and 5 sets of 3×3 matrices with coplanar columns; use `np.linalg.det()` to verify their determinants are all close to 0.

Q2 Given any 2×2 matrix A and a diagonal matrix D , explain the geometric transformation AD and DA each perform: ① $D = [1, 0; 0, 2]$; ② $D = [3, 0; 0, 1]$; ③ $D = [3, 0; 0, 2]$.

Q4 Study the basic concept of singular value decomposition (SVD) on your own.

Q6 Write a Python function `is_singular(A)` that returns whether matrix A is singular by checking whether its determinant is close to 0 (tolerance $1e-9$).

Questions Q8 – Q16

Q8 Construct 10 random pairs of 2×2 matrices A , B , and verify whether $\det(AB)$ always equals $\det(A) \cdot \det(B)$, tallying the numerical error.

Q10 Write a function `random_permutation_matrix(n)` that randomly generates an $n \times n$ permutation matrix, and verify its determinant is always $+1$ or -1 .

Q12 Explore: how does the sign of the determinant change after swapping two columns of a matrix 3 times (odd)? What about 4 times (even)? Verify your conjecture with Python.

Q14 Write a function `transpose_det_equal(A)` that verifies whether $\det(A)$ equals $\det(A.T)$, and tally the error over 10 random matrices.

Q16 Write a function `orthogonal_check(Q)` that verifies whether matrix Q satisfies $QQ^T = I$, and computes whether its determinant is ± 1 .

Q9 Write a function `verify_det_product(A, B)` that compares $\det(AB)$ with $\det(BA)$, verifying they're equal, and also checks whether AB itself equals BA .

Q11 Write a function `swap_sign_det(A, i, j)` that swaps columns i and j of matrix A , and verify whether the determinant becomes the negative of the original.

Q13 Write a function `scale_det(A, k)` that checks whether $\det(kA)$ equals $k^n \cdot \det(A)$ (n being A 's dimension), and verify with matrices of different sizes.

Q15 Use `numpy.linalg.svd` to decompose a 3×3 matrix A , verify that $|\det(U)|$ and $|\det(V)|$ are both 1, and check the sign relationship between $\det(A)$ and the product of S 's diagonal entries.

Questions Q17 – Q25

Q17 Using Python, visualize a 2×2 matrix A and its transpose A^T : plot each one's transformation of the unit square, and compare whether the two areas are equal.

Q19 Explore: given a singular matrix A (determinant 0), try computing its inverse with `numpy.linalg.inv(A)`, observe what happens, and explain why.

Q21 Explore: construct a rotation matrix R (determinant 1) and a reflection matrix F (determinant -1), compare the determinants of RF and FR , and explain the difference between rotation and reflection.

Q23 Explore the relationship between the determinant and a matrix's rank: construct square matrices of different ranks, and verify the correspondence between a zero determinant and a matrix not being full rank.

Q25 Building on this section, write a short summary (comments are fine) covering the geometric intuition behind each of these five properties: determinant 0, products, row/column swaps, scalar multiples, and transposes.

Q18 Write a function `volume_scale_factor(A, k)` that constructs the 3×3 matrix kA , and visualizes how the volume scales by k^3 as the scalar k changes.

Q20 Write a function `chain_det(matrices)` that takes a list of matrices, computes the determinant of their product, and verifies it equals the product of the individual determinants.

Q22 Write a function `visualize_ABx_vs_BAx(A, B)` that plots the final shape of the unit square under ABx and BAx , comparing their areas and shapes.

Q24 Write a function `svd_step_by_step(A)` that decomposes a 2×2 matrix A via SVD, and visualizes the shape of the unit square after each of the V^T , S , and U steps in turn.

10 Prompts to Explore Further (1/2)

- 1 Explain why collinear or coplanar column vectors lead to a determinant of 0, and how that relates to whether a matrix is invertible.
- 2 Explain the geometric meaning of $\det(AB) = \det(A) \cdot \det(B)$ — why do the two transformations' scaling factors multiply rather than add?
- 3 Explain why $AB \neq BA$ in general, yet $\det(AB) = \det(BA)$ always holds.
- 4 Explain how a permutation matrix is used to swap a matrix's rows or columns, and why its determinant is always ± 1 .
- 5 Explain why swapping any two rows or two columns of a matrix negates its determinant.

10 Prompts to Explore Further (2/2)

6

Explain the formula $\det(kA) = k^n \det(A)$ — why is the scalar k raised to the power of the matrix's dimension n ?

7

Explain why a square matrix A and its transpose A^T always have the same determinant.

8

Using the geometric view of the singular value decomposition $A = USV^T$, explain why transposing doesn't change the determinant.

9

Explain the geometric transformation each of U , S , and V represents in SVD (rotation, scaling, rotation), and how each contributes to the determinant.

10

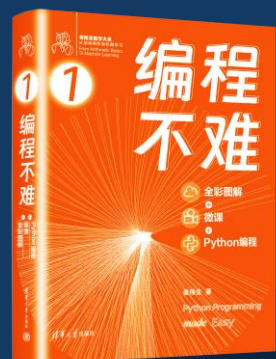
Building on this section, discuss what practical value these properties of the determinant have for testing invertibility and understanding linear transformations.

数学

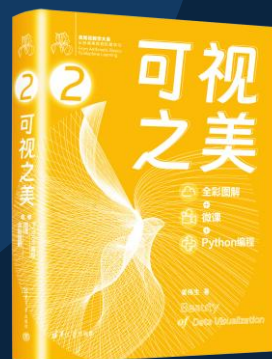
数学基础

线性代数

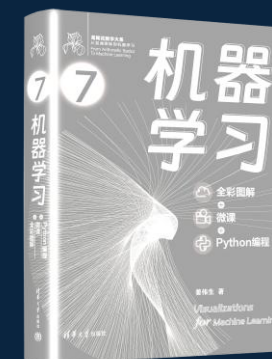
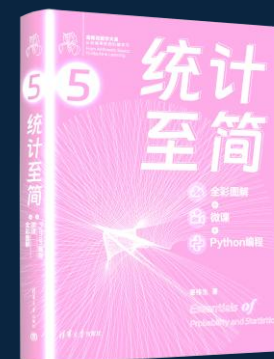
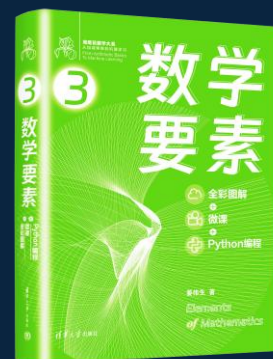
概率统计



Python编程



数据可视化



工具

数据分析
图和网络

回归、降维
分类、聚类

实践

